

RESEARCH INTEREST

- Machine Learning, Optimization Theory, Operations Research

EDUCATION

- Stanford University** California, US
 - Management Science and Engineering Sep. 2026–Jul. 2031(Expected)
- Peking University** Beijing, China
 - B.A. in Mathematics, School of Mathematical Sciences Jun. 2023–Jul. 2026
- Peking University** Beijing, China
 - B.A. in Chemistry, College of Environmental Sciences and Engineering Sep. 2022–Jun. 2023

SELECTED HONORS AND AWARDS

- Finalist in INFORMS Applied Probability Society Best Student Paper Prize, 2025**
 - Annual award recognizing outstanding student research in applied probability [Link] [Award] Oct. 2025
- Applied Mathematics Elite Program**
 - The program lead by Prof. [Weinan E](#) accepted only 15 people this year Jun. 2024
- Silver Medal, 35th Chinese Chemical Olympiad (National Final)**
 - The prize is awarded to the top 150 high school students in chemistry throughout China Nov. 2021

WORKING EXPERIENCE

- Decision Intelligence Lab, DAMO Academy** Hangzhou, China
 - Research Intern Jul. 2025–Sep. 2025
 - Developed the complete codebase for MetaFlow and built execution sandboxes (including tool calling like RAG search and Lean4 theorem proving) for online generated workflow and verification, enabling real-time execution feedback for reinforcement learning with verifiable rewards.

RESEARCH EXPERIENCE ON LARGE LANGUAGE MODELS AND OPERATIONS RESEARCH

- Online Scheduling on LLM Inference** Massachusetts Institute of Technology
 - Advisor: Prof. David Simchi-Levi, Massachusetts Institute of Technology Oct. 2024 – May. 2025
 - Optimizing LLM Inference: Fluid-Guided Online Scheduling with Memory Constraints** (α - β) [Ruicheng Ao*](#), [Gan Luo*](#), David Simchi-Levi, Xinshang Wang [SSRN] [Arxiv] [Code1] [Code2] [Demo]
 - * **Major Revision** at Operations Research.
 - * Preliminary version accepted to NeurIPS 2025 MLxOR Workshop.
 - * **Finalist in INFORMS Applied Probability Society Best Student Paper Prize, 2025.** [Link] [Award]
 - * Formulated the LLM inference as a multi-stage online scheduling task with stochastic queueable requests, proposed a novel online batching algorithm for LLM inference and proved that the algorithm achieves near-optimal throughput while controlling latency and Time to First Token (TTFT).
 - * Conducted numerical experiments on synthetic and real-world datasets with Llama-7B on A100 GPU to validate theoretical results.
- Analysis of batching and scheduling algorithms in LLM inference** Columbia University
 - Advisor: Prof. Jing Dong, Columbia University Apr. 2025 – Present
 - Work in progress, Analysis of Continuous Batching Algorithm in LLM Inference**
 - In this work, we first formulated the continuous batching algorithm as a discrete-time model. Then we analyzed its steady state and its dynamics behavior under overloaded conditions. Next we will further analyze its dynamics under admission control.
- LLM Agent and Workflow Generation** DAMO Academy
 - Advisor: Prof. Wotao Yin, DAMO Academy & Prof. Bin Dong, Peking University Apr. 2025 – Present
 - MetaFlow: A Meta Approach of Training LLMs into Generalizable Workflow Generators** (α - β) [Gan Luo*](#), [Zihan Qin*](#), Bin Dong, Wotao Yin

- * **Submitted, Under Review.**
- * Formulated workflow generation as a meta-learning problem where LLMs learn to compose task-level solution strategies from operators, producing reusable workflows that generalize across problem instances rather than instance-specific solutions.
- * Developed a two-stage training approach combining supervised fine-tuning on synthetic workflow data with reinforcement learning with verifiable rewards (RLVR), using execution feedback across instances to improve end-to-end success rates.
- * Demonstrated strong zero-shot generalization to untrained tasks and novel operator sets, achieving performance comparable to state-of-the-art baselines on in-domain tasks across benchmarks in question answering, code generation, and mathematical reasoning.

RESEARCH EXPERIENCE ON OPTIMIZATION THEORY

Convergence and Speedup Analysis of Distributed Optimization Algorithms

Peking University

Advisor: Prof. Kun Yuan, Peking University

Nov. 2023 – Jun. 2025

o Push-Pull Algorithm Provably Achieves Linear Speedup Over Arbitrary Network Topologies

Liyuan Liang*, **Gan Luo***, Kun Yuan [Arxiv] [Code: Linear Speedup] [Notes]

- * Submitted to **SIAM Journal on Optimization**.
 - * Proposed a novel multi-step descent analysis framework and first proved that the [Push-Pull algorithm](#) achieves linear speedup over arbitrary strongly connected digraphs. Our multi-step analysis resolved the non-vanishing noise issue inherent in [traditional single-step approaches](#). Also see the [\[notes\]](#).
 - * Conducted all numerical experiments to validate the linear speedup property we proved.
- #### o Achieving Linear Speedup and Optimal Complexity for Decentralized Optimization over Row-stochastic Networks
- Liyuan Liang*, Xinyi Chen*, **Gan Luo***, Kun Yuan [Arxiv] [Code]
- * **Accepted to ICML 2025, Spotlight**
 - * Introduced novel metrics to characterize the influence of row-stochastic mixing matrices and established the first convergence lower bound for decentralized optimization over row-stochastic networks.
 - * Developed a new analysis framework proving that [PULL-DIAG](#) achieves linear speedup and proposed a multi-gossip protocol that resolves instability issues and attains the lower bound with near-optimal complexity.
 - * Conducted all numerical experiments to validate the theoretical results on convergence lower bound, linear speedup, and near-optimal complexity.

SKILLS & OTHERS

- . **Computer Skills:** Python (&PyTorch), Cuda, Lean4, MATLAB, C++, LaTeX
- . **Language:** [Sichuanese dialects](#) (Native), Mandarin Chinese (Native), English (Fluent)
- . **The Test of English as a Foreign Language (TOEFL-IBT):** Total 101 with 27(R)+28(L)+22(S)+24(W)
- . **Conference Reviewing:** ICLR 2026